HDOC Day-35

Matrix Operations

Question: Write a menu-driven C program using separate functions for the operations shown in the assignment (options 1-3 and 4 to exit).

C Program:

```
#include <stdio.h>
#define MAX 20

void distinct(int a[MAX][MAX], int n) {
    int i, j;
    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (a[i][i] == a[j][j]) {
                printf("Diagonal elements are not distinct.\n");
                return;
            }
        }
    }
    printf("All diagonal elements are distinct.\n");
}

void diagonalSum(int a[MAX][MAX], int n) {
    int i, sum = 0;
    for (i = 0; i < n; i++)
        sum += a[i][i];
    printf("Sum of main diagonal = %d\n", sum);
}

void traverse(int a[MAX][MAX], int n) {
    int d, i, j;
    printf("Diagonal traversal: ");
    for (d = 0; d <= 2 * (n - 1); d++) {
        if (d % 2 == 0) {
            for (i = n - 1; i >= 0; i--) {
                j = d - i;
                if (j >= 0 && j < n)
                    printf("%d ", a[i][j]);
            }
        } else {
            for (i = 0; i < n; i++) {
                j = d - i;
                if (j >= 0 && j < n)
                    printf("%d ", a[i][j]);
            }
        }
    }
    printf("\n");
}

int main() {
    int a[MAX][MAX], n, i, j, choice;
    printf("Enter order of square matrix (1-20): ");
    scanf("%d", &n);
    if (n < 1 || n > MAX) return 0;
    printf("Enter matrix elements:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    do {
        printf("\n1. Check distinct diagonal elements");
```

```
        printf("\n2. Sum of main diagonal");
        printf("\n3. Diagonal traversal");
        printf("\n4. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: distinct(a, n); break;
            case 2: diagonalSum(a, n); break;
            case 3: traverse(a, n); break;
            case 4: printf("Exit\n"); break;
            default: printf("Invalid choice\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Output: For matrix 1 2 3 / 4 5 6 / 7 8 9: diagonal elements are distinct; sum = 15; traversal = 1 2 4 7 5 3 6 8 9.